

Characteristics of wind flow around a target building with different surrounding building layers predicted by CFD simulation

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ABSTRACT

This study explored appropriate surrounding building regions that can ensure an accurate prediction of wind flow characteristics around the target building. A validated computational fluid dynamics (CFD) program solving the Reynolds-Averaged Navier Stokes (RANS) equations with $k-\epsilon$ turbulence model is used to investigate the influences of surrounding buildings with different heights, densities and arrangements on the target building. An empirical formula is developed to determine an appropriate spatial domain in simulation for different combinations of building density and height under varying layout modes. The proposed rules are further demonstrated and verified by simulating the wind flow of a target building in Dalian, China. Simulation results confirm that the surrounding building height, density and layout have a considerable impact on the simulated core region and single influencing factor cannot determine a reasonable flow simulation domain. The developed guide can effectively predict the wind flow conditions around buildings with an acceptable modeling accuracy and effort.

1. Introduction

Airflow around buildings in an urban area is a critical consideration in the design of building layouts, natural ventilation, and human comfort [1,2]. The airflow around buildings can be determined from wind tunnel tests [3–5], site tests [6,7] and simulations based on Computational Fluid Dynamics (CFD) [8,9]. With the development of computer technology and computational models, the accuracy and speed of CFD have improved considerably, and an increasing number of researchers have used CFD to calculate urban ventilation.

The use of CFD to simulate airflow in urban configurations can be classified into three categories: full-scale models, mesoscale models (10–200 km), and individual buildings. Full-scale model is highly accurate in simulating airflow, but it is very time-consuming to establish all the details of buildings within the area under consideration, and the calculations require large computational resources [10,11]. By contrast, an individual building model may lead to inaccurate simulation results. Hooff et al. [12] simulated the natural ventilation of a large semi-enclosed stadium and found that there is a substantial gap between the results of the model with or without surrounding buildings. Kim et al. [13] and Quan et al. [14] demonstrated that buildings surrounding a target building have a significant effect on the simulation results.

Chang et al. [15] compared the pressure on the roof of an isolated building and that when it is surrounded by multiple buildings, and results showed substantial differences. Therefore, a reasonable modeling range can ensure the accuracy of results and reduce computational time.

Appropriate modeling within a computational range is necessary for obtaining accurate results [16]. For the study of urban airflow fields, it is essential to clearly describe the interactions between urban morphology and computational range. Tominaga et al. [16] suggested modeling surrounding buildings within a radius of $1H-2H$ from the target building, where H is the height of the target building. Furthermore, at least one extra block in each direction is replicated around the simulated area. Tong et al. [17] suggested simulation ranges based on the aspect ratios of buildings. Three layers of buildings are built around the target building when $H/W = 1$, where H and W are the height of the building and the width of the canyon, respectively. Influence Region can be reduced to two layers for wide canyons ($H/W = 1/3$) and high-rise configuration. Sumei Liu et al. [18] suggested modeling surrounding buildings within a radius of $3R$ from the target building, where R is the maximum dimension of the target building, and pointed out that the main factors affecting air flow around the target building are its immediate surroundings. Lee [19] suggested modeling surrounding buildings within a radius of 200 m in the city of Copenhagen. It is

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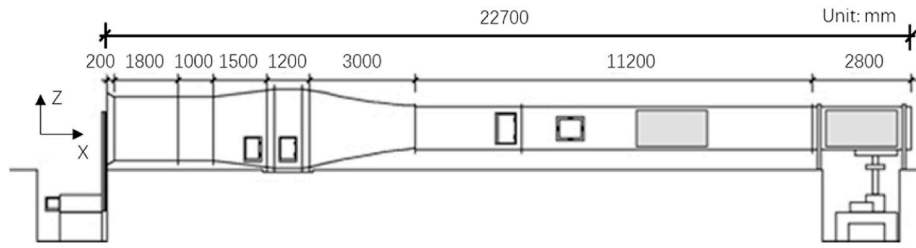


Fig. 1. Structure of the wind tunnel.

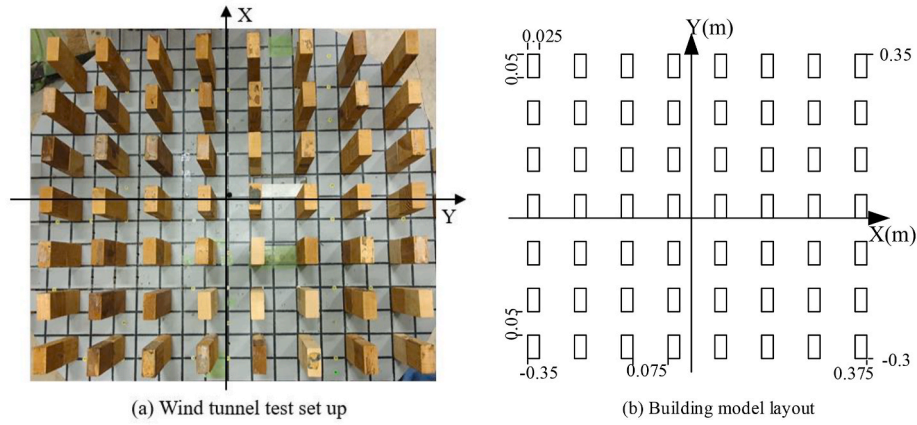


Fig. 2. A simple city block model.

concluded that surrounding buildings within a radius of 200 m around the target building should be included. The modeling range of buildings around the target building is related to many factors, such as building density, building height, and building layout [20,21]. Liu et al. [22] pointed out that the aspect ratio has a significant effect on air flow. Buccolieri found that air flow around buildings is strongly depended on building density [23]. Chang et al. [24] and Hang et al. [25,26] remarked that the arrangement and height of buildings affect urban ventilation.

In previous studies, the factors considered in the relationship between simulation range and building height, density, and layout are not comprehensive enough. This article will further refine these building parameters for a more comprehensive simulation. In this paper, wind tunnel test results have been used to obtain an appropriate simulation model to determine air flow around a target building. In this study, several cases have been simulated to determine a suitable equation for calculating the simulation ranges for different urban morphologies. Comparing the simulation results of different modeling ranges around the target building in Dalian, the correctness of the equation is verified. The purpose of the study is to reduce the workload of modeling and to reduce computational costs while maintaining an acceptable range of accuracy in determining air flow fields.

2. Validation of CFD method and model

Firstly, the accuracy of the simulation model is verified by wind tunnel tests. The experiments are taken in an open-circuit low-speed wind tunnel designed for wind environmental assessment and ventilation studies at the School of Architecture and Wind Engineering, Tokyo Polytechnic University. The wind tunnel has a test section of 1.2 m wide, 1.2 m high and 22.7 m long (show as Fig. 1). A simple city block model is used for wind tunnel tests as show in Fig. 2. The size of each block is: width $W = 25$ mm (x direction), length $L = 50$ mm (y direction), height $H_w = 100$ mm (z direction). The uncertainty of measurements is conducted according to ISO GUM [27]. First, the accuracy of the test instrument split film probe (SFP) is measured under the setting angle $\alpha = 0^\circ$ and 90° . These relative expanded uncertainties are approximately 3%. Then the extended uncertainty of wind velocity around a building is measured. Their relative expanded uncertainty is less than 10% [28].

2.1. Computational domain

This study uses the Reynolds-Averaged Navier-Stokes (RANS) equations with a $k-\epsilon$ turbulence model to solve for the turbulent airflow parameters in the computational domain. RANS equations with a realizable $k-\epsilon$ turbulence model are widely used in the simulation of urban

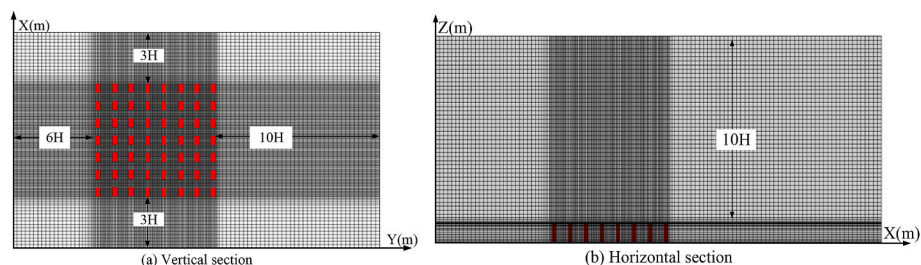


Fig. 3. Computational domain and grid discretization.

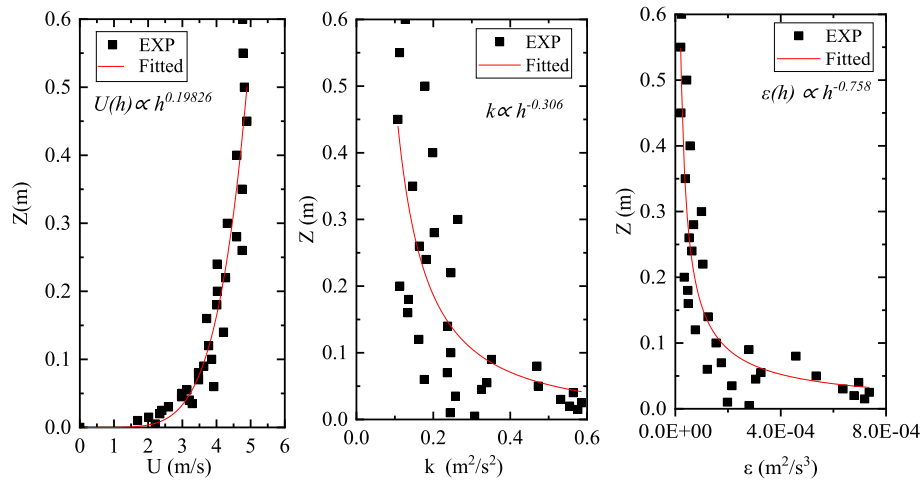


Fig. 4. Inflow boundary conditions.

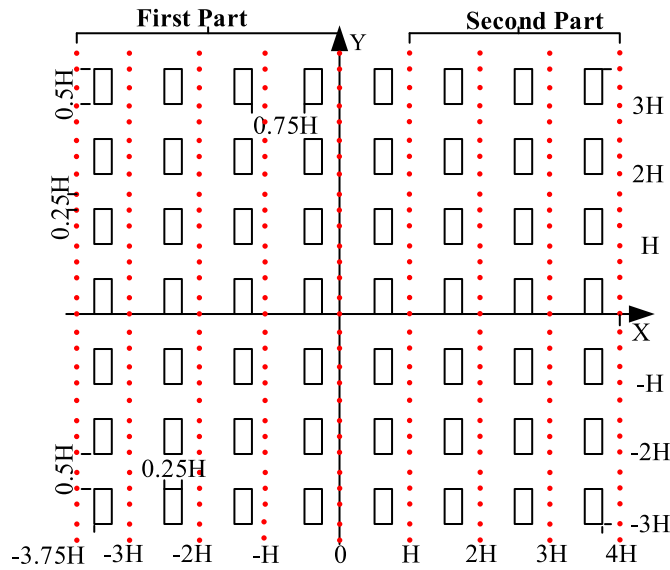


Fig. 5. Simple city block model and measuring point locations.

ventilation with high accuracy [29–31]. The computational domain of the buildings has the same cross-section as the wind tunnel in the vertical (z) and lateral (y) directions. The distance between the boundaries and the domain top, domain inlet, and domain outlet are $10H_w$, $6H_w$, and $10H_w$, respectively.

2.2. Grid discretization

Grid sensitivity studies are carried out to ensure that the results of the selected grid will not change significantly with finer grid resolution. The validation studies used fine grid (minimal grid size is 0.002 m, 4.09 million cells in total), middle grid (minimal grid size is 0.005 m, 2.78 million cells in total) and coarse grid (minimal grid size is 0.00625 m, 1.81 million cells in total). The cell size is increased by a factor of 1.2 in a direction away from the building. Grid convergence study is performed by using the grid convergence index (GCI) [32]. The GCI is described as $GCI = F_S |\varepsilon| / (r^p - 1)$, where $F_S = 1.25$ is the safety factor, ε is the relative error between the results of two grid solutions that need to be compared, and $p = 2$ represents a second-order method. The variable r is the ratio of the cell sizes of two meshes, i.e., the refined and the original meshes. The validation study used the fine grid with a GCI value of 2%. Therefore, the fine grid is used in the research. Fig. 3 shows the grid discretization.

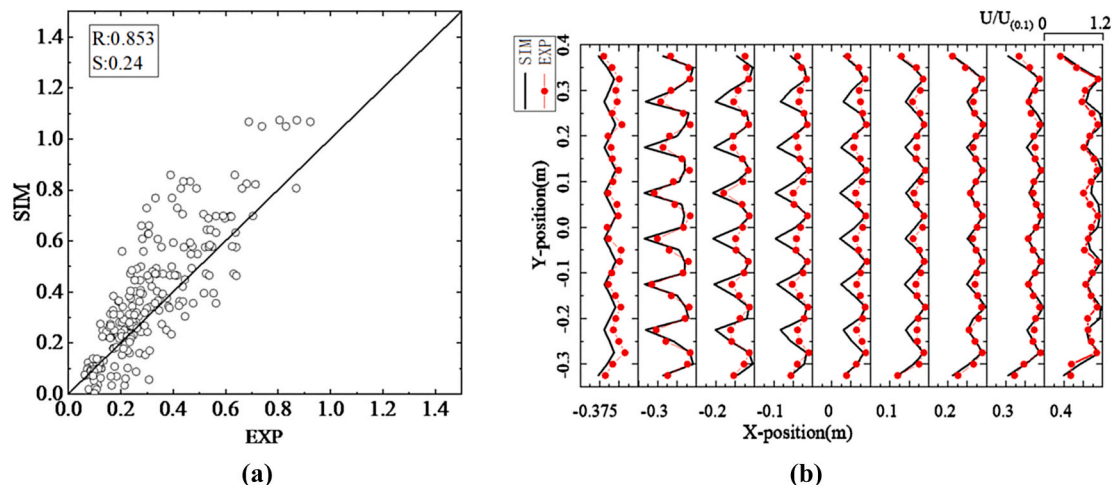


Fig. 6. Correlation of experimental and calculation results and comparison of wind speed ratios at measurement points.

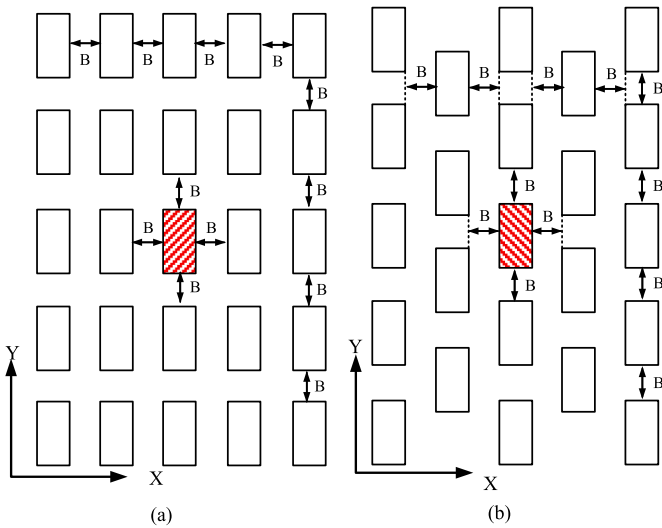


Fig. 7. Schematic of building models: (a) PA; (b) SA.

2.3. Boundary conditions

The inlet profiles of U , k , and ε are plotted in Fig. 4. The inlet average flow velocity and variable flow velocity in the wind tunnel is measured at different heights at a longitudinal direction (x) of -0.4 m. The functions representing the variations of the turbulent kinetic energy (k) and turbulent dissipation rate (ε) versus distance along the z -direction are obtained. A power-law function is used to obtain these relationships. Outflow boundary condition is specified at the end of the domain. Lateral and upper surfaces of the computational domain of the computational domain are considered as wall. A standard wall function is used for the wall surfaces. The value of y^+ is within the range of 30–300 for all cases.

2.4. Validation

To verify the accuracy of the model, the simulation results are compared with experimental results. Fig. 5 shows the measuring point distribution at $0.5H_w$ in the middle of the buildings. The experimental data and simulation results are compared in Fig. 6. The correlation coefficient and standard error are denoted as R and S , respectively. In general, the simulated flow velocity ratios are higher than those obtained from wind tunnel tests. The valley values of the wind tunnel experiments are in good agreement with the simulated values. The overall RMSE between the experimental values and the simulated values is 13%, which met the requirement of the ASHRAE Guideline 2014 [33].

The RANS equations with a realizable k - ε turbulence model are verified using the results of the wind tunnel tests, and good agreement is obtained. Therefore, the model is used for subsequent calculations.

3. CFD based parametric study

3.1. Case description

To obtain an appropriate simulation range around a target building, a series of buildings with different heights, densities, and arrangements are simulated using the RANS equations with a realizable k - ε turbulence model.

The model is composed of uniformly arranged blocks with a cross-sectional shape of $25\text{ m} \times 12.5\text{ m}$, and building heights of 25 m (H), 50 m ($2H$), 75 m ($3H$), and 100 m ($4H$). The target building is located at the center of these buildings. Airflow is monitored for a parallel arrangement (PA in Fig. 7-(a)) and a staggered arrangement (SA in Fig. 7-(b)). The spacings between buildings (B) are 7.5 m , 12.5 m , 25 m ,

Table 1

The relationship between building spacing and building density.

B(m)	7.5	12.5	25	50	75
Building density	45%	30%	14%	6%	3%

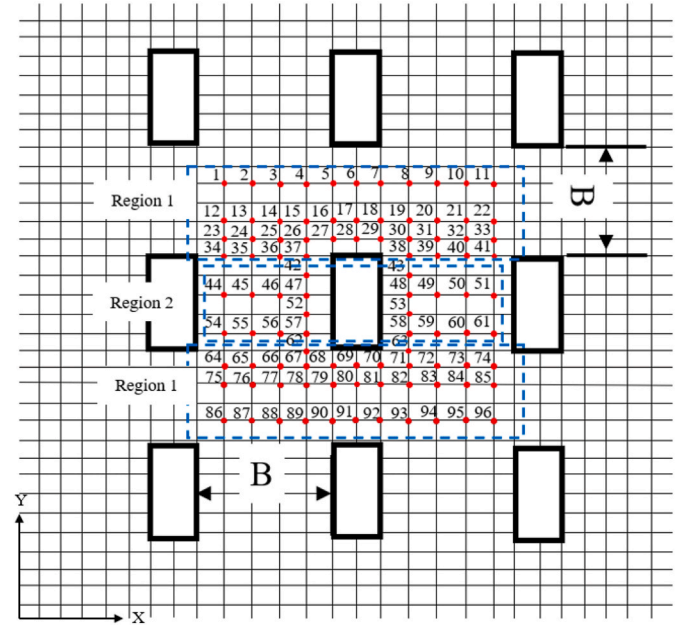


Fig. 8. Monitoring points.

50 m , and 75 m . For this study, ‘building density’ is defined as the ratio of the area covered by the buildings to the building lot area. Thus, different building spacings corresponds to different building densities. The relationship between building spacing (B) and building density (D) is shown in Table 1.

There are 40 cases in total which are composed of 4 groups of building heights (H , $2H$, $3H$, $4H$), 5 groups of building density (45%, 30%, 14%, 6%, 3%), and two types of building arrangements (PA, SA). In each case, the wind velocity around the target building with a different number of surrounding layers is compared. Fig. 7 depicts the PA and SA arrangements for two surrounding layers around the target building.

Air flow around the target building is simulated with the same boundary conditions as wind tunnel simulation. The inlet air flow direction is in longitudinal direction (X). Airflow is deemed as stable when velocity around buildings no longer varies noticeably with increasing number of surrounding layers.

3.2. Results and analysis

3.2.1. Comparisons of air flow with different layers

In this study, three heights are selected to arrange monitoring points. The height of 2.5 m above the ground (Low) is considered to be the typical height for human activity. Airflow is closely related to human comfort and safety in this area. At the middle of the building height (Middle), the airflow around the target building is related to the natural ventilation of most floors. Airflow at a height of 2.5 m above building roofs (High) can be used to judge the wind resistance level and safety of buildings. The distribution of monitoring points at Low is shown in Fig. 8. Other height monitoring point distribution is identical with Low. The monitoring points are divided into two regions. Region 1 is located on the left and right sides of the building, and Region 2 is located on the windward and leeward sides of the target building.

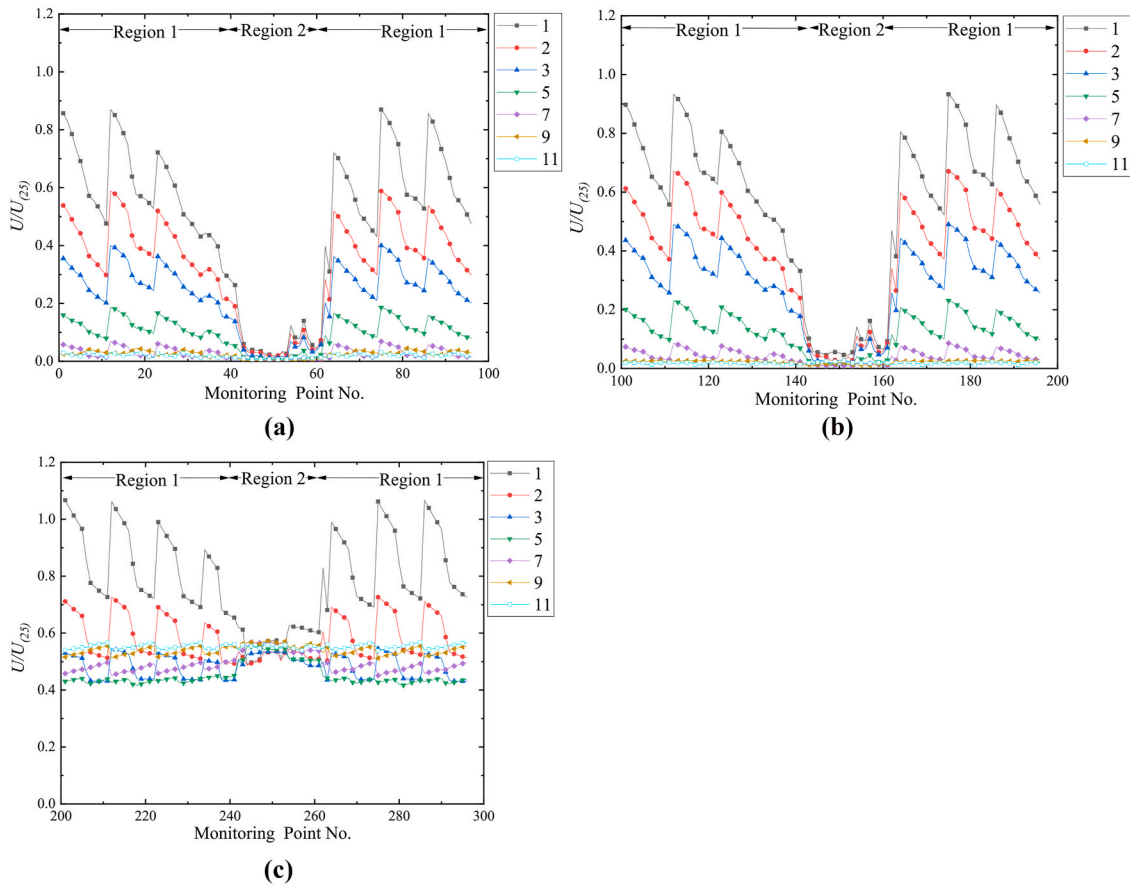


Fig. 9. Velocity ratios with different surrounding layers around the target building when the building density in the PA mode is 45%, and the building height is 25 m (The different lines in the picture represent different surrounding layers around the building): (a) Velocity ratios at Low; (b) Velocity ratios at Middle (c) Velocity ratios at High.

First, the number of surrounding layers required for the parallel arrangement of buildings is analyzed. Airflow velocities of monitoring points with different number of surrounding layers are compared. Fig. 9 shows flow velocity ratios when the building density in the PA mode is 45%, and the building height is 25 m. Flow velocity ratios are defined as the ratios of the scalar velocities near the ground ($z = 2.5 \text{ m} = H_{(25)}/10$, Low), middle of the building height ($z = 25 \text{ m} = H_{(25)}/2$, Middle), and 2.5 m above the top of the building ($z = 27.5 \text{ m} = 1.1 \times H_{(25)}$, High) to the inflow velocity ($H = 25 \text{ m}$, $U_{(25)} = 3.12 \text{ m/s}$) at the top of building.

According to the simulation results, the number of surrounding layers required for the velocity to stabilize in Region 1 and Region 2 is different when buildings are in parallel arrangement. The velocity of Region 1 needs more surrounding layers to achieve stability when the number of surrounding layers around the target building increased. Since computing costs are high when the velocity remains constant in Region 1 and urban configurations rarely have such long parallel arrangements, this study uses Region 2 for analysis.

Fig. 10 shows the velocity ratio of the monitoring points in Region 2 with the increase of surrounding building layers. The velocity declines substantially with the increasing of the sheltering effect. Transitioning from an isolated building configuration to the configuration surrounded by surrounding layers, the velocity declines significantly due to the addition of upwind buildings. The effect becomes negligible near the target building, and velocity reaches a convergence when there are five surrounding layers (see Fig. 10 -(a)), five surrounding layers (see Fig. 10 -(b)), and seven surrounding layers (see Fig. 10 -(c)) around the target building, at the Low, Middle, and High positions, respectively. Therefore, to simulate the wind environment of buildings, the Low and Middle need at least five layers while High needs seven layers.

Subsequently, the number of surrounding layers required for a staggered arrangement of buildings is analyzed. Fig. 11 compares the velocity ratios when the building density in the SA mode is 45%, and the building height is 25 m. The flow velocities in Region 1 and Region 2 have an identical decrease tendency. Hence, Region 1 and Region 2 are considered as the criteria for evaluating the needed surrounding layers. Due to eddy currents between buildings, the flow velocities at the top of the buildings fluctuate. Therefore, it is difficult to determine the needed surrounding layers at the High position. Hence, the number of surrounding layers that need to be modeled around the target building at High position is not analyzed in this paper.

Fig. 12 shows the calculated velocity ratio of the monitoring points in Region 1 and Region 2 as the surrounding layers increase. The surrounding buildings will affect the airflow around the target building. The effect becomes negligible near the target building, and velocity reaches a convergence when there are three surrounding layers (see Fig. 12 -(a)), and three surrounding layers (see Fig. 12 -(b)) at the Low and Middle positions, respectively. Hence, three surrounding layers are necessary to simulate the wind environment of buildings at the Low and Middle positions.

3.2.2. Layers required for simulation

According to the above evaluation criteria, the number of surrounding layers required for different building arrangements is determined. Table 2 and Table 3 present the number of layers required to simulate different building patterns.

Table 2 shows the number of surrounding layers that needed to be simulated in PA arrangement. With an increase in building height, the number of surrounding layers of buildings simulated at the Low, Middle,

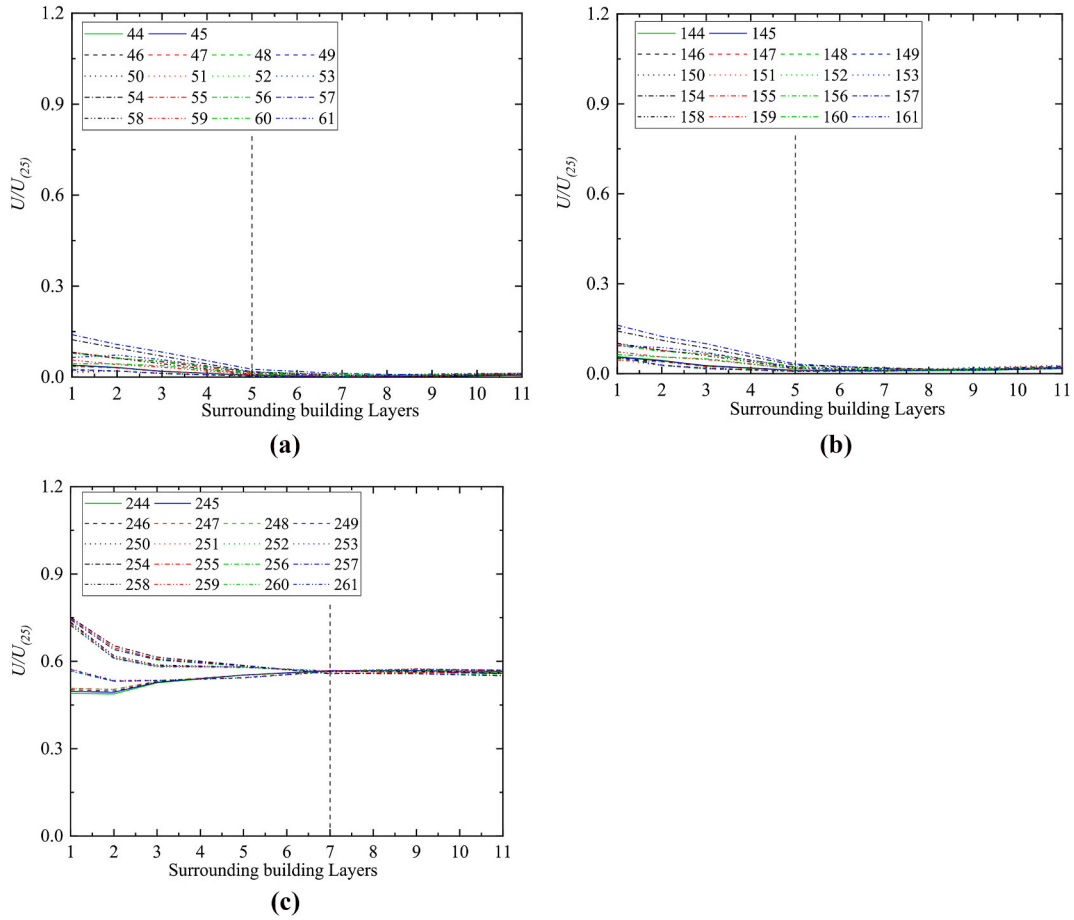


Fig. 10. Convergence chart of velocity ratios with different surrounding layers around the target building when the building density in the PA mode is 45%, and the building height is 25 m (The different lines in the picture represent the velocity ratio variation of the different monitoring point): (a) Velocity ratios at Low; (b) Velocity ratios at Middle (c) Velocity ratios at High.

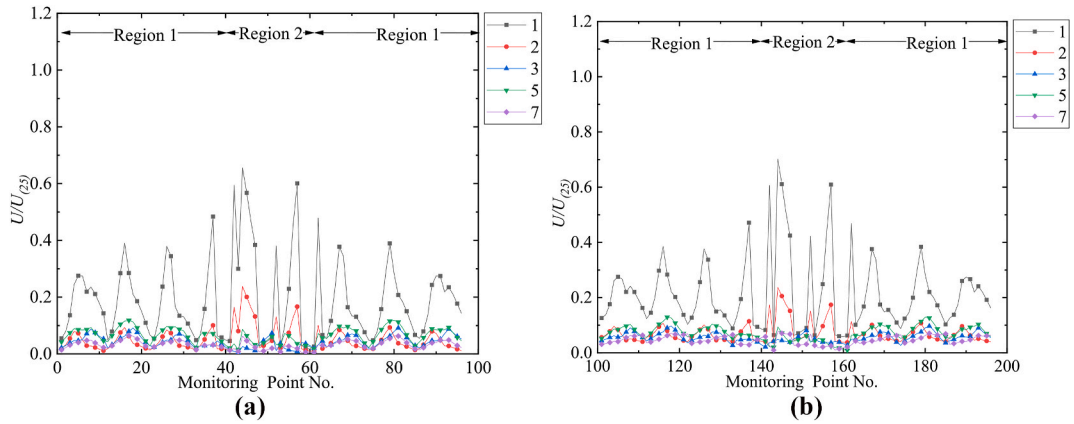


Fig. 11. Velocity ratios for different layers around target building when building density in the SA mode is 45%, and the building height is 25 m. (The different lines in the picture represent different layers around the building): (a) Velocity ratios at Low; (b) Velocity ratios at Middle.

and High positions increased. For higher building densities, more buildings needed to be simulated, as shown in Fig. 13.

The fitting equation for the number of surrounding layers to be simulated around the target building is as follows:

$$L_p = a + b \times Z + c \times D \quad (1)$$

where L_p is the number of layers to be simulated in PA arrangement, Z (m) is the building height, and D (%) is building density. a , b , and c are

constants ($a = 0.879$, $b = 0.065$, and $c = 0.078$, $R^2 = 0.862$ for the Low; $a = 0.761$, $b = 0.066$, and $c = 0.073$, $R^2 = 0.851$ for the Middle; $a = 3.8$, $b = 0.053$, and $c = 0.051$, $R^2 = 0.832$ for the High).

The coefficient of significance for each variable is less than 0.05, which is statistically significant and met the requirement of the ASHRAE Guideline 2014 that R^2 should be greater than 0.8 [33].

Table 3 shows the number of layers that need to be simulated for a staggered arrangement of buildings.

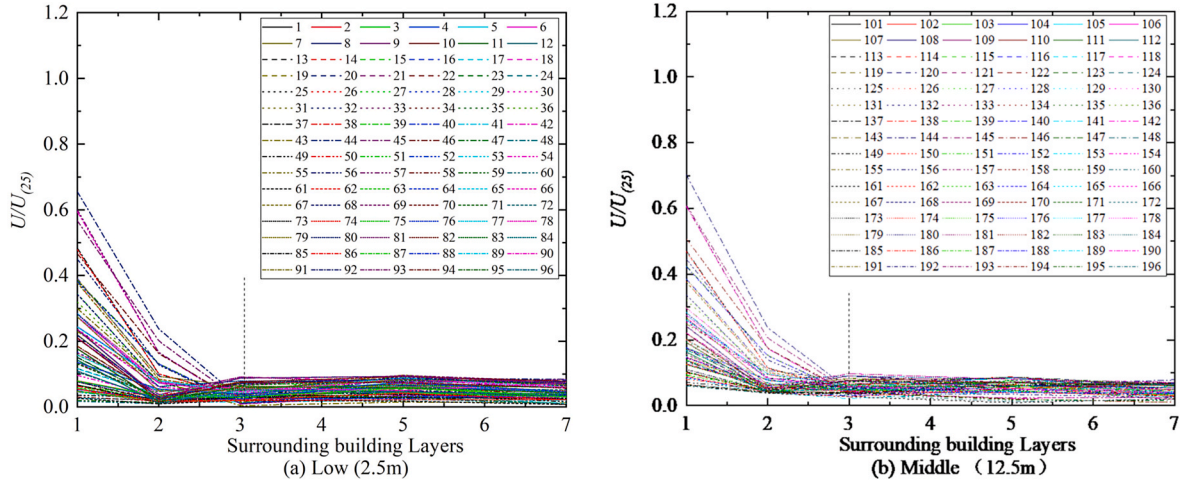


Fig. 12. Convergence chart of velocity ratios with different surrounding layers around the target building when the building density in the SA mode is 45%, and the building height is 25 m (The different lines in the picture represent the velocity variation of the different monitoring point): (a) Velocity ratios at Low; (b) Velocity ratios at Middle.

Table 2

The number of layers required for the CFD simulation of a parallel arrangement of buildings.

Building height (m)	Density Position	45%	30%	14%	6%	3%
25	Low	5	5	5	3	2
	Middle	5	5	5	3	2
	High	7	7	5	5	5
50	Low	7	7	5	5	3
	Middle	7	5	5	5	3
	High	9	9	9	7	7
75	Low	9	9	9	7	5
	Middle	9	9	9	7	5
	High	9	9	9	9	7
100	Low	11	9	9	7	7
	Middle	11	9	9	7	7
	High	11	11	11	9	9

Table 3

The number of surrounding layers required for CFD simulation in SA arrangement.

Building height (m)	Density Position	45%	30%	14%	6%	3%
25	Low	3	5	5	5	7
	Middle	3	5	5	5	7
50	Low	5	7	7	7	9
	Middle	5	7	7	7	9
75	Low	7	7	9	9	9
	Middle	5	7	7	9	9
100	Low	7	7	9	9	11
	Middle	5	7	9	9	11

The number of surrounding layers simulated at the Low and Middle positions are dependent on building height and building density, as shown in Fig. 14. The number of surrounding layers have a negative correlation with building density and have a positive correlation with building height. The fitting equation for the number of surrounding layers around the target building is as follows:

$$L_s = a + b \times Z + c \times D \quad (2)$$

where L_s is the number of layers simulated in SA arrangement, Z (m) is the building height, and D (%) is building density. a , b , and c are constants ($a = 5.755$, $b = 0.046$, and $c = -0.079$, $R^2 = 0.857$, for the Low; a

$= 6.111$, $b = 0.040$, and $c = -0.087$, $R^2 = 0.831$, for the Middle).

4. Demonstration and verification

To verify the accuracy of the simulation range obtained in this study, airflow distribution around the target buildings in Dalian city is simulated. The staggered arrangement is similar to the arrangement of the community. The target building has a height of 54 m, a length of 32 m, and a width of 9 m, and the building density is 15%.

4.1. Boundary conditions

The weather data are obtained from the meteorological station in Dalian. The power-law model is used for the vertical velocity profile of the inflow boundary, and the vertical profiles for k_z and ε_z are obtained from Richards and Hoxey [34]:

$$U_z = U_r \times \left(\frac{z}{z_r}\right)^\alpha \quad (3)$$

$$k_z = U_{ABL}^{*2} / \sqrt{C_\mu} \quad (4)$$

$$\varepsilon_z = U_{ABL}^{*3} / \kappa(z + z_0) \quad (5)$$

where U_r (m/s) is the velocity at a reference height z_r (m), z_0 (m) is the rough wall height, and z (m) is height. The average wind speed (U_r) at the weather station is 4.2 m/s and the height (z_r) of the weather station is 10 m above the ground. Since the meteorological station is located in an urban area, the exponent used in the power-law is $\alpha = 0.33$ [35]. κ is the Karman constant ($\kappa=0.4$) and U_{ABL}^* is the atmospheric boundary layer friction velocity. U_{ABL}^* is calculated from the velocity U_r (m/s) at reference height z_r (m) as:

$$U_{ABL}^* = \kappa U_r / \ln((h + z_0) / z_0) \quad (6)$$

Outflow boundary condition is specified at the end of the domain. Lateral and upper surfaces of the computational domain of the computational domain are considered as wall.

4.2. Geometric model

The downstream and both sides buildings have little influence on the airflow around the target building [17]. Therefore, the simulation area mainly considers the building density and the number of surrounding

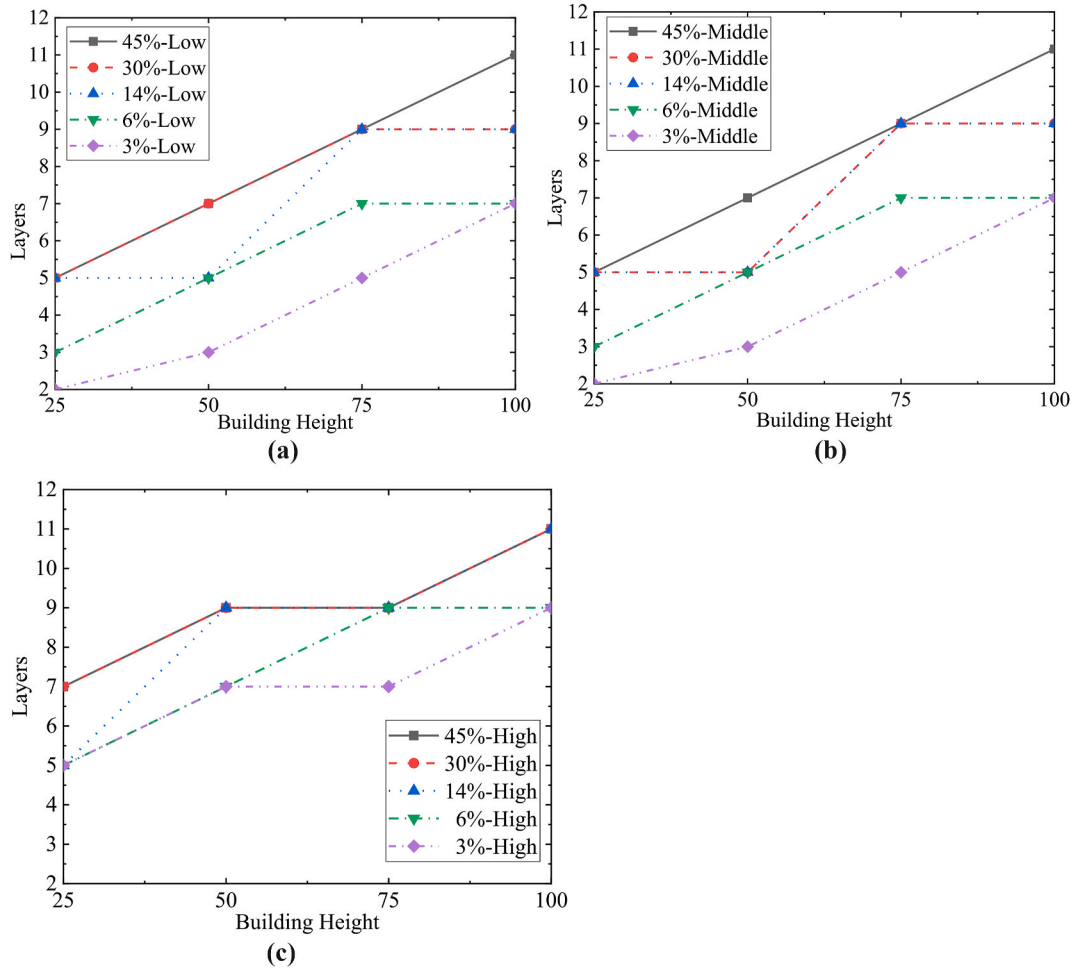


Fig. 13. Surrounding layers needed to be simulated at different locations for a parallel arrangement of buildings.

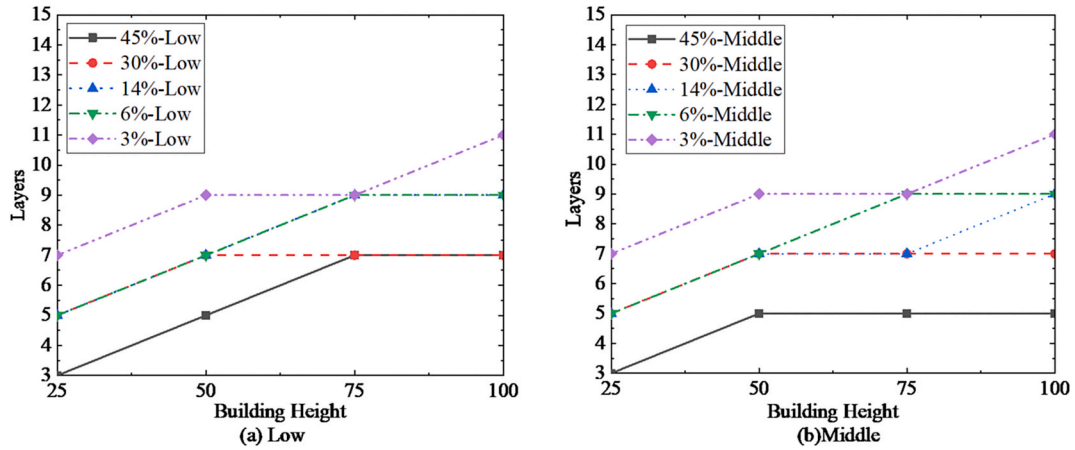


Fig. 14. Surrounding layers needed to be simulated at different locations in SA arrangement.

layers on the windward side. Based on the equation obtained from the above simulation, seven surrounding layers of buildings should be built around the target building for a building height of 54 m and a building density of 15% in SA arrangement (see Eq.(2)).

In this study, the maximum scope of the simulation region is extended to nine layers, and the overall modeling area is shown in Fig. 15.

The validation models, Vcase0 to Vcase5, have 9 layers, 7 layers, 5

layers, 3 layers, 2 layers, and 1 layer each. In this study, a tetrahedral mesh mixing scheme is adopted, due to its easy adaptability to geometric structures. To simplify the model, the components of the buildings are ignored if the maximum dimension of the component is less than 2 m. The maximum grid size of the target building and other buildings are 0.5 m and 8 m, respectively. According to the recommendations of CFD guidelines [36], the distances from the city boundary to the top, inlet, and outlet of buildings are $9H$, $5H$, and $20H$ respectively, where H is the

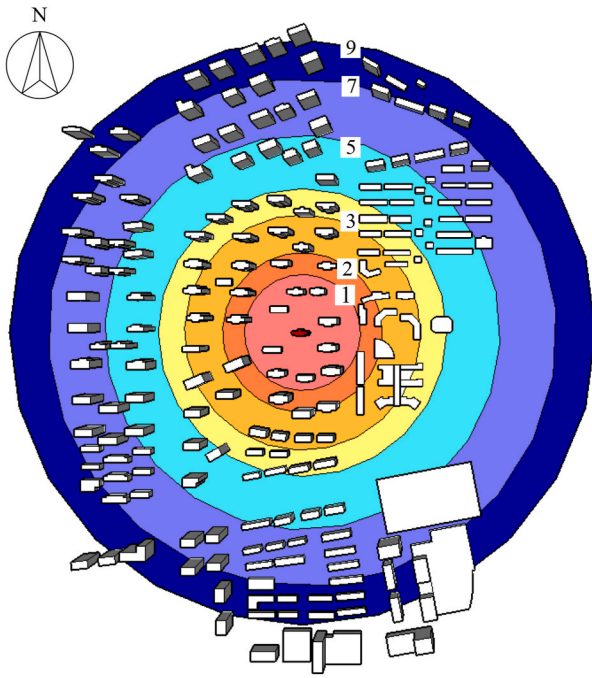


Fig. 15. Multiple layers of buildings around the target building.

height of the highest building. The grid mesh size is 17.14 million, 15.5 million, 14 million, 12.8 million, 10.5 million, and 4.5 million cells for Vcase0 to Vcase5, respectively.

4.3. Results

In this study, the airflow around the target building obtained by Vcase0 and other simplified models is compared, as shown in Fig. 16.

Fig. 16-(a), Fig. 16-(b), and Fig. 16-(c) depict the airflow at the Low, Middle, and High positions, respectively. In most areas, the simplified model has a higher velocity than the detailed model.

Airflow in Vcase 0 to Vcase 2 enclosed by the bold lines is similar in most areas and quite different from Vcase3 to Vcase5 because the surrounding environments of Vcase3 to Vcase5 are excluded. Table 4 presents the values of average air velocity for the urban configuration areas of interest and the relative error between Vcase0 and the other cases.

The relative error between Vcase0 and Vcase1 is about 20%. Relative error increased with the simplification of the model. Due to the shielding effect between buildings, the detailed model has a lower average velocity. The lower right velocity around the target building of Vcase1 is located in the recirculation area, and its average velocity is greater than that of Vcase 0 and Vcase 2.

Further investigations of the monitoring point velocities around the target building are conducted. The distribution of monitoring points is shown in Fig. 17. Vcase0, Vcase1, and Vcase2 show similar trends and wind speed value (see Fig. 18). It is observed that there are fewer

Table 4
Average velocity and relative error.

	Vcase 0	Vcase1	Vcase2	Vcase3	Vcase4	Vcase5
Velocity(m/s)	0.9	0.729	0.578	1.15	2.216	2.366
(Low)rowhead						
Relative errorrowhead	0	19%	35%	28%	146%	162%
Velocity(m/s)	0.575	0.689	0.65	1.39	1.81	2.52
(Middle) rowhead						
Relative errorrowhead	0	19.8%	13%	141%	214%	338%
Velocity(m/s)	0.809	0.98	0.777	1.81	3.2	3.26
(High)rowhead						
Relative errorrowhead	0	21%	4%	123%	219.5%	302%

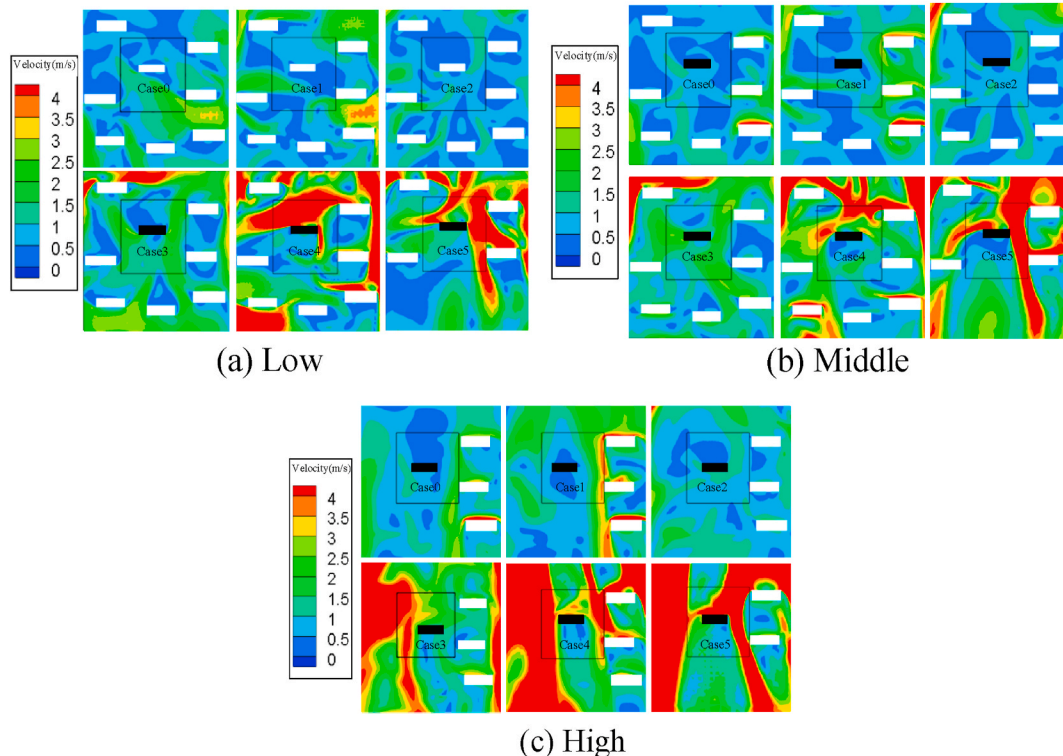


Fig. 16. Air flow around the target building.

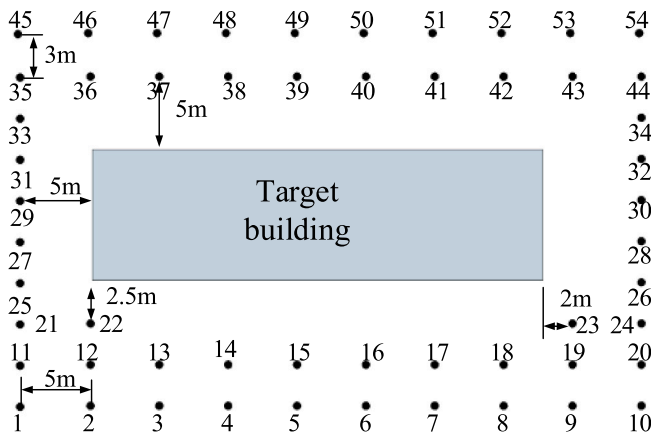


Fig. 17. Monitoring points around target building.

surrounding layers around the target building, and the velocity around the building has a higher velocity. Greater velocity differences are observed at the higher position.

The relative errors of average velocity in Vcase0 and other cases are presented in Table 5. The relative errors of Vcase1 and Vcase2 are similar. The relative error increases with the simplification of the building model. The relative error of Vcase1 and Vcase0 is about 20%, which is similar to the relative error of Vcase2 and Vcase0.

This paper further compares the simulated values with tested values. The test is in December 28, 2019, and the dominant wind at the test site is northward wind. Four corresponding measuring points are arranged around the target building. Point 1 is 2.5 m higher than the roof of the target building. Point 2 and Point 3 are 0.8 m and 1.8 m away from the wall at the middle height of the building. Point 4 is 1.5 m above the roof of the highest building in the community (see Fig. 19). Table 6 lists the specifications of the main test instruments. Flow velocities at Point 1 and Point 4 are measured by a three-dimensional anemometer while those at Point 2 and Point 3 are measured by a hot-wire anemometer.

Fig. 20: depicts airflow around the target building at different testing points. Since the data from the weather station is the average wind speed over a 2-min period, the corresponding time at the measuring points is compared.

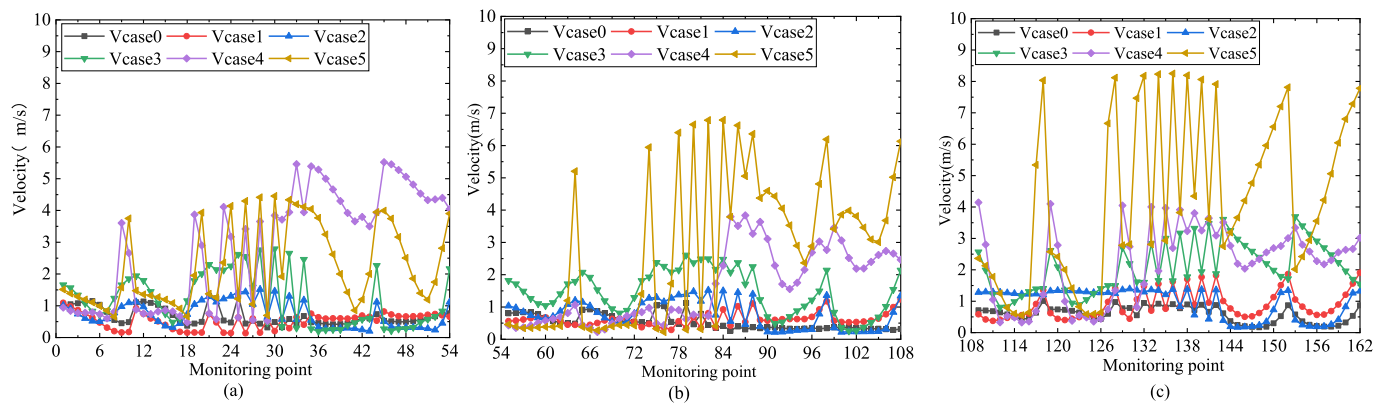


Fig. 18. Wind speed at each monitoring point: (a) wind speed at Low; (b) wind speed at Middle; (c) wind speed at High

Table 5
Relative errors of Vcase0 and other cases.

		Vcase0	Vcase1	Vcase2	Vcase3	Vcase4	Vcase5
Relative Error (%)	Low	0	16%	16%	39%	58%	29%
	Middle	0	14%	22%	46%	86%	43%
	High	0	20%	8%	53%	60%	48%

The velocity trend at Point 1 is similar in the Y and Z directions. The velocity at Point 4 has the similar trend in the X and Z directions. The airflow velocities of different measuring points are shown in Table 7. Point 2 and Point 3 are located at 0.8 m and 1.8 m from the wall respectively. The speed at Point 3 is higher than that at Point 2. Point 1 and Point 4 are at a higher location. Hence the wind speeds and fluctuations are higher.

Fig. 21 compares the differences between the measured and simulated velocities. The simulation results of Vcase0, and Vcase1 are close to the test values. Several measurement points of Vcase2, Vcase3, Vcase4 and Vcase5 are significantly different from the measured values.

The relative errors of the experimental and simulation data are depicted in Table 8. The average relative errors based on the experimental data for Vcase0 to Vcase5 are 23.6%, 22.6%, 132.8%, 282.5%, 106.2%, and 348.5%, respectively. There are differences between the geometric model and the actual building structure. Due to the complexity of the cases, we consider that the calculation results of Vcase0 and Vcase1 are acceptable.

This paper compared the average speed and the speed of the monitoring points around the target building when the target building has different number of surrounding layers around it. And the speed of the test is compared with the simulation results. We can draw a conclusion that 7 surrounding layers are the appropriate simulation range. In conclusion, the appropriate simulation range obtained from equation (2) deduced from previous studies is reasonable.

5. Conclusions

In this study, an appropriate CFD model is used to simulate a series of cases consisting of different building heights, densities, and layouts. For the CFD simulations, the appropriate number of surrounding layers around the target building is obtained using the formulation presented above. Thereafter, a portion of the urban configuration of Dalian city is simulated, and the airflow field around the target building is simulated when there are different surrounding layers around it. The study led to the following conclusions.

- (1) With the increase number of surrounding layers around the target building, the velocity around the target building will converged. There is a significant difference between the airflow around a



Fig. 19. Distribution of measuring points in the community.

Table 6

The specifications of the main test instruments.

Instruments	Measurement range	Accuracy
THREE-DIMENSIONAL	0-50 (m/s)	±1.5%
Hot-wire anemometer	0-30 m/s	±3%

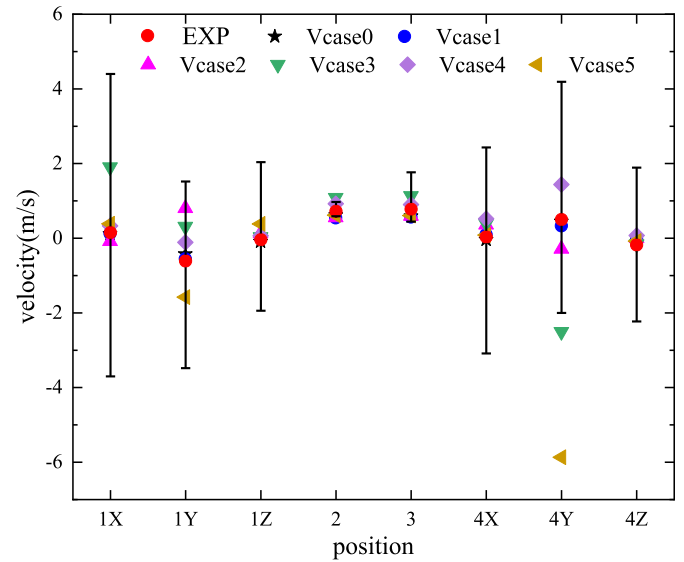


Fig. 21. Comparison of experimental and simulation data.

Table 8

Relative error of Experimental and simulated values.

		Vcase0	Vcase1	Vcase2	Vcase3	Vcase4	Vcase5
Relative Error (%)	1X	28	19	158	1167	118	155
	1Y	28	7	233	151	81	164
	1Z	31	29	279	143	188	638
	2	15	25	24	50	28	15
	3	25	27	24	46	16	22
	4X	17	7	92	92	95	64
	4Y	11	35	159	602	187	1675
	4Z	34	32	94	9	137	55

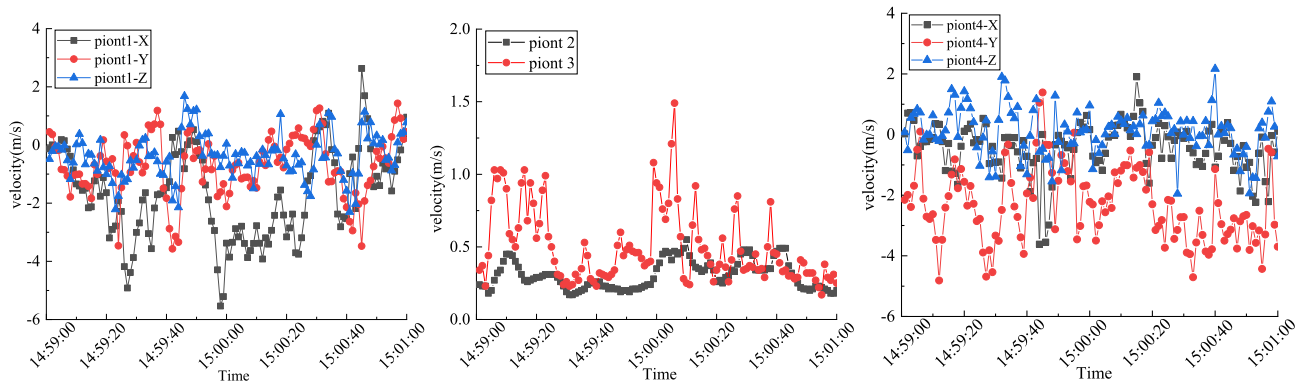


Fig. 20. Measured values of the air flow field around buildings.

Table 7

Airflow velocity at different measuring points.

	Velocity at Point 1 (m/s)			Velocity at Point 2 (m/s)	Velocity at Point 3 (m/s)	Velocity at Point 4 (m/s)		
	X	Y	Z			X	Y	Z
Average	0.15	-0.611	-0.0702	0.72	0.775	0.033	0.5	-0.179
MAX	4.4	1.519	2.0398	0.97	1.765	2.433	4.19	1.891
MIN	-3.7	-3.481	-1.9402	0.59	0.445	-3.087	-2	-2.229
Mode	0.2	-0.911	-0.0402	0.62	0.775	0.033	0.1	-0.279

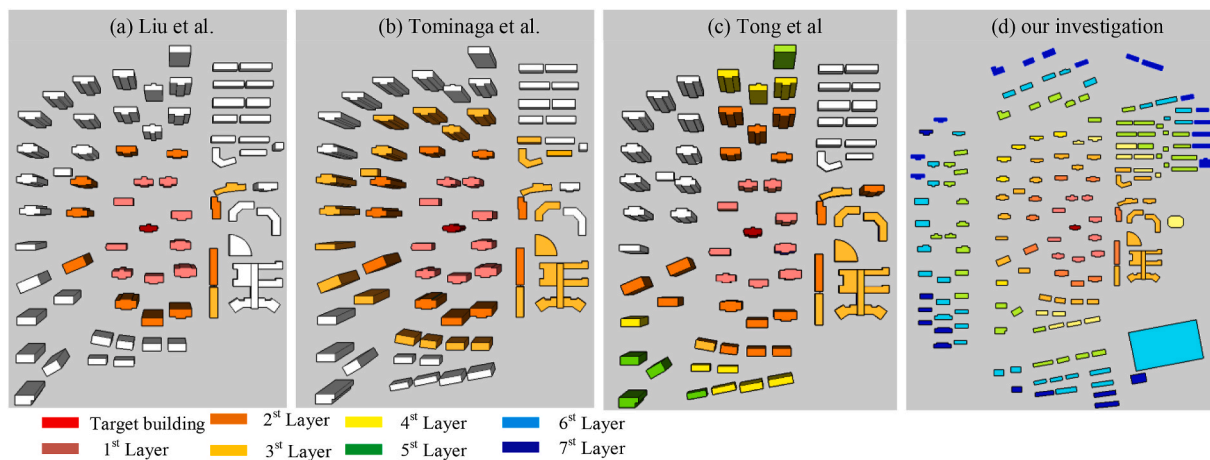


Fig. 22. Comparisons of the regions where surrounding buildings are constructed according to: (a) Liu et al. [18], (b) Tominaga et al. [16], (c) Tong et al. [17], and (d) our investigation.

single target building and the steady airflow around the target building.

- (2) The accuracy of simulation results is ensured and the number of grids can be reduced when an appropriate simulation range is used. The appropriate range can reduce calculation times and the work involved in constructing a geometric model.
- (3) The appropriate simulation range varies based on building height, density, and arrangement. Multiple factors should be considered for determining the simulation range.
- (4) An equation for calculating the appropriate simulation range is determined which can be used as a practical guide for simulating air flow around buildings.

This paper further compares the regions based on data available in the literature. Liu et al. [18] have presented the smallest region (see Fig. 22-(a)). They suggest using a region three times the maximum dimension of the target building as a scale in the radial direction to represent surrounding buildings. Tominaga et al. [16] and Tong et al. [17] designated ranges that increased in different directions. Tominaga et al. [16] suggested using twice the height of the target building and an additional street block as the simulation radius (see Fig. 22- (b)). Tong et al. [17] suggested using three layers of buildings around the target building as the simulation range (see Fig. 22-(c)). The simulation range obtained in this study is larger than each of the previous studies presented. The results of this investigation may therefore be more comprehensive. However, the model has several limitations:

- (1) Due to the complexity of the real urban configuration, the arrangement and height of the buildings are not regular. Hence, there may be some degree of deviation from the results of the equation considered in this study.
- (2) The application of the theory may also be affected if there are large differences in the surrounding building densities along different directions around the target building.
- (3) The effect of trees and other obstacles are not considered in this study, hence the surrounding layers of CFD simulation near the ground may be reduced.
- (4) The building model for validation has been simplified. The building structure less than 2 m around the target building is ignored in the model. Therefore, the results of the model may be different from real building.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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